

Exam Date & Time: 08-May-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Examination - April / May -2024

Design Patterns [CSE-4065]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) List the main essential elements of the design pattern using UML diagram. (2)
 - A)
 - B) Apply Model view controller architecture pattern in case of designing library management software. (3)
 - C) Distinguish between creational and structural design patterns using any suitable example with respect to intent, problem, applicability, solution and meta UML model. (5)
- 2) List out the intent, problem, solution and meta UML model with suitable example for builder design pattern. (5)
 - A)
 - B) Apply Builder design pattern and draw solution in the form of UML diagram for the scenario to convert a rich text format (RTF) document into different representations like Text, ACSII and HTML. (5)
- 3) List out the intent, problem, solution and meta UML model with suitable example for facade design pattern. (5)
 - A)
 - B) Apply decorator design pattern for designing a notification library for third part application with SMS, email and Facebook notifications. (5)
- 4) List out the intent, problem, solution and meta UML model with suitable example for Template design pattern. (5)
 - A)
 - B) Apply and draw meta model of Iterator design pattern with suitable example. (5)

- 5) Apply singleton design pattern for the following scenario:
Draw the solution in the form of UML diagram for the scenario for maintaining log information of application which helps developer to check logs, errors, and debugging messages of applications. (5)
- A)
- B) Apply Strategy design pattern for the following scenario:
Consider a scenario to sort the list of integers using three different algorithms such as Quick sort, Insertion sort and Merge sort. Sort strategy can be selected optionally during run time based on time and space complexities. Draw solution in the form UML diagram by applying strategy design pattern. (5)

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